

# Probability - Set 1

TMUA practice questions by topic. Specification M7.1-M7.7. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 A bag contains 4 red and 6 blue counters. Two counters are taken at random without replacement. Find the probability that they are different colours.

- A  $\frac{24}{100}$   
B  $\frac{15}{25}$   
C  $\frac{5}{8}$   
D  $\frac{14}{16}$   
E  $\frac{12}{15}$   
F  $\frac{12}{25}$

- 2 Two fair dice are thrown. Find the probability that the product of the two scores is odd.

- A  $\frac{2}{3}$   
B  $\frac{4}{9}$   
C  $\frac{1}{4}$   
D  $\frac{1}{6}$   
E  $\frac{2}{3}$   
F  $\frac{1}{3}$

- 3  $P(A) = 0.6$ ,  $P(B) = 0.5$  and  $P(A \text{ and } B) = 0.3$ . Which of the following statements is/are true?

- I  $A$  and  $B$  are independent.  
II  $A$  and  $B$  are mutually exclusive.  
III  $P(A \text{ or } B) = 0.8$

- A none of them  
B I only  
C II only  
D III only  
E I and II only  
F I and III only  
G II and III only  
H I, II and III

- 4 A spinner has probability 0.15 of landing on red. It is spun 200 times. What is the expected number of times it lands on red?
- A 30  
B 40  
C 15  
D 35  
E 25  
F 20
- 5 In a class of 30, 18 study French, 12 study German and 5 study both. A student is chosen at random. Given that the student studies French, what is the probability that they also study German?
- A  $\frac{1}{6}$   
B  $\frac{2}{5}$   
C  $\frac{1}{8}$   
D  $\frac{1}{9}$   
E  $\frac{1}{8}$   
F  $\frac{1}{12}$
- 6 Three fair coins are tossed. Find the probability of getting more heads than tails.
- A  $\frac{3}{4}$   
B  $\frac{1}{4}$   
C  $\frac{1}{8}$   
D  $\frac{3}{8}$   
E  $\frac{1}{8}$   
F  $\frac{1}{2}$

- 7 The probabilities that a train is early, on time or late are 0.1,  $x$  and  $2x$  respectively. Find the probability that the train is not late.
- A 0.5  
 B 0.9  
 C 0.7  
 D 0.4  
 E 0.6  
 F 0.3
- 8 A fair die is thrown repeatedly until a six appears. Find the probability that the first six appears on the third throw.
- A  $\frac{25}{216}$   
 B  $\frac{36}{216}$   
 C  $\frac{216}{216}$   
 D  $\frac{25}{216}$   
 E  $\frac{216}{216}$   
 F  $\frac{216}{216}$
- 9 Two cards are drawn at random from a standard pack of 52 without replacement. Which of the following expressions is the probability that both are aces?
- A  $\frac{4}{52} + \frac{3}{51}$   
 B  $\frac{4}{52} \times \frac{3}{52}$   
 C  $\frac{4}{52} \times \frac{3}{51}$   
 D  $\frac{4}{52} \times \frac{3}{52}$   
 E  $\frac{4}{52} \times \frac{3}{51}$   
 F  $\frac{4}{52} \times \frac{3}{52}$

- 10** Events  $A$  and  $B$  are mutually exclusive with  $P(A) = 0.3$  and  $P(B) = 0.4$ . Find  $P(\text{neither } A \text{ nor } B)$ .
- A** 0.3  
**B** 0.7  
**C** 0.1  
**D** 0.58  
**E** 0.12  
**F** 0.42
- 11** A biased coin has probability  $p$  of landing heads. It is tossed twice. The probability of exactly one head is  $\frac{3}{8}$ . Find the possible values of  $p$ .
- A**  $\frac{1}{4}$  or  $\frac{2}{3}$   
**B**  $\frac{1}{4}$  only  
**C**  $\frac{1}{8}$  or  $\frac{7}{8}$   
**D**  $\frac{1}{4}$  or  $\frac{3}{4}$   
**E**  $\frac{3}{8}$  only  
**F**  $\frac{3}{16}$  only
- 12** A box contains  $n$  white balls and  $3$  black balls. Two balls are drawn without replacement. The probability that both are white is  $\frac{5}{12}$ . Find  $n$ .
- A** 6  
**B** 4  
**C** 5  
**D** 3  
**E** 7  
**F** 9